

C_CPE_2601

**SAP Certified - Backend Developer
- SAP Cloud Application
Programming Model**

SAP CERTIFICATION TASKS

Course Version: 2601

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Typographic Conventions

American English is the standard used in this handbook.

The following typographic conventions are also used.

This information is displayed in the instructor's presentation	
Demonstration	
Procedure	
Warning or Caution	
Hint	
Related or Additional Information	
Facilitated Discussion	
User interface control	<i>Example text</i>
Window title	<i>Example text</i>

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Introduction

In this assessment, you will work on one or more exercises that you must complete in the assigned system. Please make sure to read the instructions below carefully before accessing the system.

Before starting:

- Ensure your browser is up to date and clear your browser cache and cookies.
- Make sure to close all browser sessions before you start the assessment and remember not to open unnecessary tabs or windows while working on the exercises.

During the assessment:

- Ensure you follow the assessment instructions precisely. Any deviation may result in failure of the exercise.
- Avoid access to any other practical systems during the assessment in any other browser session (including incognito).
- If you accidentally close the assessment browser, go back to learning.sap.com, log in again, and go back to the assessment.
- Should you encounter technical difficulties with the assessment, please refer to the [FAQs](#) or [create a case](#) with the support team.

Ending the assessment

- Select the *Confirm completion* button to submit your assessment. This button is in the browser tab where you initially accessed the assessment, and not in the system tab.
- Once you submit the assessment, you can no longer make changes. Do not attempt to access the system again once you have submitted your assessment.
- To get the result of your assessment, please wait for the validation process to complete (approximately 15 minutes). Your result (failed/passed) will be displayed under *My Certifications*. Click on your profile icon in the top-right corner of learning.sap.com and select *My Certifications*.

Good luck with your assessment!

Context

You have been tasked with building a Logistics Management System for a global shipping provider. The company needs to calculate shipping fees dynamically based on the weight of packages and the mode of transport. To ensure the system is performant, the database should only store raw data, while the Total Weight and Shipping Fee must be calculated on-the-fly when the data is requested via OData. The system will eventually be deployed to SAP Business Technology Platform (BTP) using SAP HANA as the persistence layer.

Technical Constraints & Requirements

- **Framework:** SAP Cloud Application Programming Model (CAP) Node.js.
- **Persistence:** Local development uses SQLite; Production must use **SAP HANA**.
- **Security:** For the purpose of this exam, authentication is disabled (auth: none).
- **Deployment:** The application must be packaged as an **Multi-Target Application (MTA)** and deployed to **SAP BTP, Cloud Foundry runtime**.

Prerequisites

- Access the SAP Business Application Studio
- Create and open a dev space of type "Full Stack Cloud Application".
- Clone the submission template from GitHub <https://github.com/sap-samples/btp-developer-training-exam>
- Ensure to `clean-install` the dependencies
- Continue with Task 1

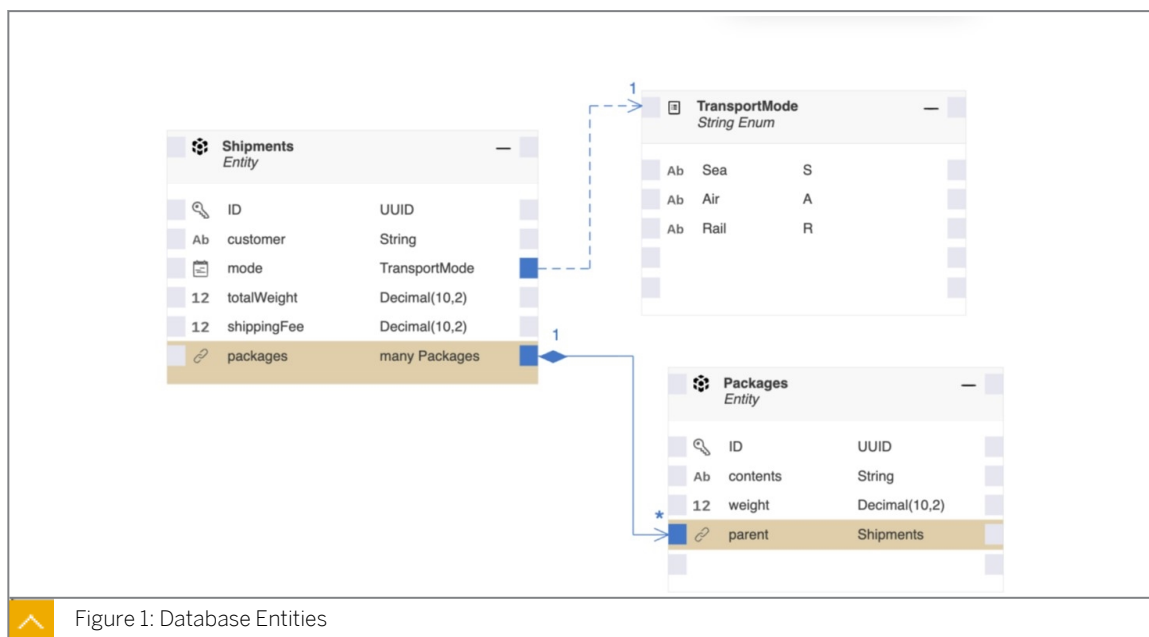
Unit 1

Task 1

Data Modeling and Data initialization

Define the data model in the db/ folder based on the following requirements. Please pay special attention to the name and order of fields as well as the data types.

- **Namespace:** exam.logistics
- **Enum:** Create a type TransportMode with keys A (Air), S (Sea), and R (Rail).
- **Entity Shipments:** Includes a unique ID, a customer name, and a transport mode. It must hold a composition of many packages.
- **Entity Packages:** Includes a unique ID, a description of the contents, and the weight (Decimal).
- **Virtual Fields:** The Shipments entity must include two virtual fields: totalWeight and shippingFee.



Mock Data

Initialize your database with mock data. You must create the necessary folders and files manually following standard CAP naming conventions. Hint: The entity uses UUIDs, while the mock data does not. The system will still process this successfully.

Table 1: Shipments

ID	Customer	Mode
AA-123	Alpha Corp	Air
BB-456	Beta Ltd	Sea

Table 2: Packages for shipment AA-123

ID	contents	weight
P1	Package 1	10
P2	Package 2	5.5

Table 3: Packages for shipment BB-456

ID	contents	weight
P3	Package 3	100

Service Exposure and Service Logic

Expose the entities via a service named `LogisticsService` in the `srv/` folder.

- The service should be reachable at the path `/logistics`.
- Ensure that both `Shipments` and `Packages` are accessible, but `Packages` should be read-only.
- 1. Implement a functional logic handler in a `.js` file matching your service definition.
 - **Requirement:** When reading `Shipments`, calculate the virtual fields.
 - **Calculation Logic:**
 - a. `totalWeight`: The sum of the weight of all packages associated with that shipment.
 - b. `shippingFee`: The `totalWeight` multiplied by a rate determined by the `TransportMode`:
 - i. **Air (A)**: 15
 - ii. **Sea (S)**: 5
 - iii. **Rail (R)**: 8
 - **Hint:** Your logic must handle cases where the packages might not be expanded in the initial request.

Production Readiness and Deployment

Prepare the application for the SAP BTP Cloud Foundry environment..

1. SAP HANA Configuration

Add the necessary dependencies and configuration to use SAP HANA in production.

2. MTA Packaging

Generate a deployment manifest. Ensure it is configured for a Node.js runtime and an HDI container, without XSUAA.

3. Set application route manually in deployment descriptor

Please overwrite the generated parameters entry in `mta.yaml` for the `<service>-srv` module and set it to the following:

```
parameters:
  instances: 1
  buildpack: nodejs_buildpack
  app-name: ${space}-c_cpe_submission-srv
  routes:
    - route: ${org}-cpe-submission.${default-domain}
```

4. Build and Deploy

Use the Cloud MTA Build Tool and the CF CLI to deploy the archive into the `SUBMISSION` space of the provided SAP BTP subaccount.

5. Submission

Please verify that your service is running and deployment was successful via the Cloud Foundry CLI in the SAP Business Application Studio.

Once you have verified your deployment, you can submit your practical exam.